

HEALTHPLUS US MEMBER SEGMENTATION & RETENTION ANALYSIS

BUSINESS QUESTION

Which HealthPlus members are most likely to lapse their health insurance, what drives their decision to leave, and which targeted retention interventions by member segment will generate the highest return on retention investment?

EXECUTIVE SUMMARY

HealthPlus loses 24.2% of its 500,000 policyholders annually - 120,837 members - representing \$732.5 million in lifetime value lost per cohort.

Major Findings

- Premium price increases are the strongest lapse driver. Members who did not renew faced average increases of 8.2% versus 5.6% for members who stayed - a statistically significant difference across all plan tiers (see Exhibit A).
- Members with two or more policies lapsed at 15.2% versus 27.0% for single-policy members - an 11.8 percentage point gap and the largest retention difference found (see Exhibit B).
- Three signals in HealthPlus records identify high-risk members before they cancel: requesting a competitor quote (33.9% lapse rate vs 22.9% average), filing a complaint (33.8% vs 23.7%), and downgrading coverage (31.1% vs 23.3%). Each raises lapse risk by at least 7.8 percentage points (see Exhibit C).

Recommendations for Action

- Flag members facing premium increases above 8% for a personal retention call before renewal.
- Target the 193,125 Potential Loyalist members currently lapsing at 26.3% with a multi-policy bundling offer. Members with two or more policies lapsed at 15.1% versus 27.0% for single-policy members.
- Enroll all new members in automatic payment at sign-up. Members without it lapsed at 29.9% versus 21.6% enrolled - an 8.3 percentage point reduction at no additional cost.

Analytical Overview

Data from 500,000 members were cleaned then analyzed using descriptive statistics, seven hypothesis tests, customer segmentation into five groups, churn classification model, CLV regression, and a time series forecast. The churn classification model distinguished members who stayed from those who left with a score of 0.663 out of 1.0. The CLV regression model explained 72.9% of the variation in member lifetime value.

Exhibit A — Average Premium Increase: Members Who Lapsed vs Members Who Stayed

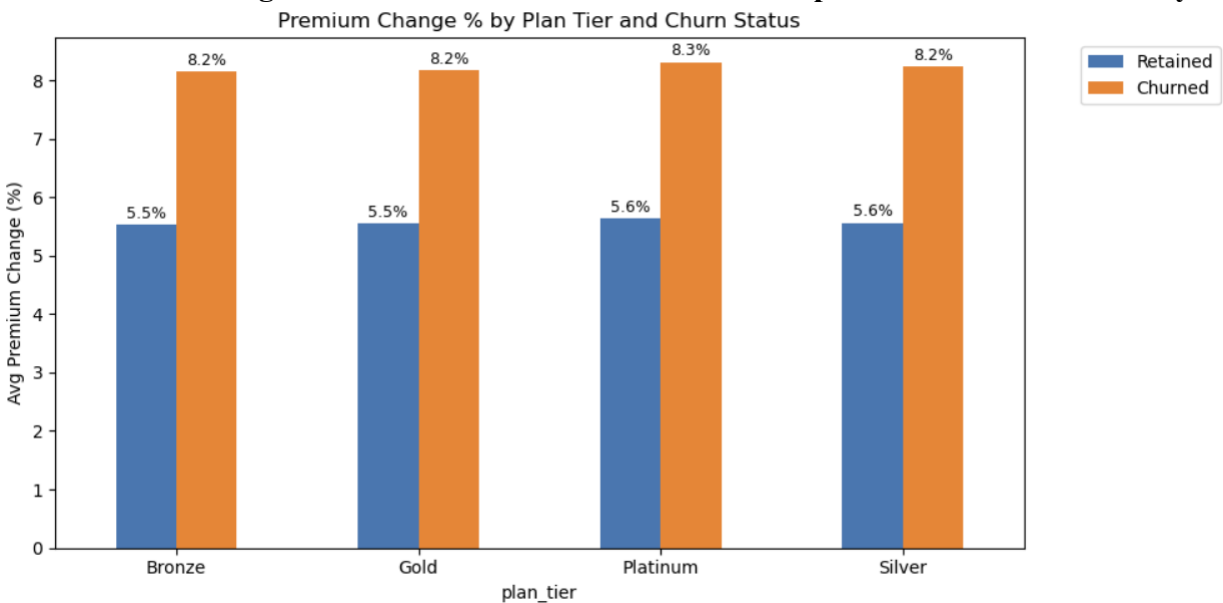


Exhibit B — Annual Lapse Rate by Policy Count

	count	churn_rate%	pct%
multi_policy_flag			
0	380009	27.0	76.0
1	119991	15.2	24.0

Exhibit C — Lapse Rate by Warning Signal vs No Signal

	count	churn_rate%	pct%
complaint_flag			
0	477242	23.7	95.4
1	22758	33.4	4.6
quote_requested_flag			
0	444207	22.9	88.8
1	55793	33.9	11.2
coverage_downgrade_flag			
0	447372	23.3	89.5
1	52628	31.2	10.5

COMPREHENSIVE ANALYSIS

DATASET OVERVIEW

The analysis uses the HealthPlus US MemberChurn 2025 dataset that represents the complete policyholder base of HealthPlus Insurance Group as of December 31, 2025. There are 501,500

rows and 47 columns. The dataset included demographic, financial, behavioral, and policy-related variables.

Attribute	Value
Raw records	501,500 rows · 47 columns
Post-cleaning	500,000 rows · 48 columns (1 derived: high claim indicator)
Geography	50 US States + DC · 4 Census Regions
Plan tiers	Bronze · Silver · Gold · Platinum (ACA age-rated premiums)
Policy types	Health HMO · Health PPO · Medicaid Supplement · Medicare Supplement · Short-term · Life · Dental/Vision
Target variable	Lapse flag: 0 = Retained (75.9%) · 1 = Lapsed (24.1%)
Lapse rate	24.1% - 120,703 lapsed · 379,297 retained
Average lifetime value	\$6,233 per member · range \$500 to \$48,575

DATA CLEANING SUMMARY

Five categories of data quality issues were identified and resolved before any analysis was conducted. All cleaning was performed in the Data Cleaning notebook using a systematic detect-then-treat approach for each issue type. Outliers were detected using visual inspections by plotting boxplots and histograms, descriptive statistics, IQR method, Z-score method, and domain bound checks.

Issue Log:

Column Name	Issue Description	Magnitude	Resolution
education_level	Missing value: education level.	6.00%	Imputed using mode.
employment_status	Missing value: employment status (blank and N/A)	4.03%	Imputed using mode.
annual_income	Missing value: annual income.	4.99%	Imputed using median income grouped by employment_status.
credit_score	Missing value: credit score.	3.99%	Imputed using median credit score grouped by age_band.
monthly_premium	Missing value: monthly premium.	0.80%	Imputed using median premium grouped by plan_tier.
net_monthly_premium	Missing value: net monthly premium.	0.80%	Derived as monthly_premium minus subsidy_amount_monthly, with a minimum value of \$0.

coverage_amount	Missing value: coverage amount.	1.80%	Imputed using median coverage amount grouped by policy_type.
rx_monthly_cost	Missing value: rx_monthly_cost.	2.50%	Filled with 0 (assumes no prescription drug costs for that period)
total_claim_amount_12m	Missing value: total claim amount.	1.50%	Filled with 0 (assumes no settled claims yet).
digital_engagement_score	Missing value: digital_engagement_score.	3.00%	Imputed using overall median.
nps_score	Missing value: nps_score.	7.00%	Filled with "No Response" category.
churn_type	Missing value: churn_type (N/A).	0.02%	Filled with "Other/Unknown" category.
annual_income	Outlier: Placeholder value 999,999 entered by staff when income is unknown at enrolment	0.09%	Treated as missing; re-imputed using median grouped by employment_status.
annual_income	Outlier: Extreme high values above \$450,000 - mix of legitimate ultra-high earners and data entry errors	0.09%	Capped at 99th percentile.
monthly_premium	Outlier: Values below \$10 including \$0.01 - system placeholder written when premium not yet calculated	0.05%	Treated as missing; re-imputed using median grouped by plan_tier.
monthly_premium	Outlier: Values above \$4,800 — keyboard ×10 entry error (e.g. \$9,240 entered instead of \$924)	0.04%	Capped at 99th percentile.
credit_score	Outlier: Value of 0 - system default written when credit score was not retrieved	0.05%	Treated as missing; re-imputed using median grouped by age_band.
credit_score	Outlier: Values between 1–299 - below the minimum possible FICO score of 300	0.05%	Clipped to lower bound of 300.
credit_score	Outlier: Values between 851–998 - above the maximum possible FICO score of 850	0.05%	Clipped to upper bound of 850.
coverage_amount	Outlier: Value of \$0 - policy type not yet mapped to a coverage amount in the system	0.05%	Treated as missing; re-imputed using median grouped by policy_type.
coverage_amount	Outlier: Value of \$999,999,999 - system sentinel placeholder for	0.05%	Treated as missing; re-imputed using median grouped by policy_type.

	unlimited or unmapped coverage		
total_claim_amount_12m	Outlier: Negative values - payment reversals recorded with incorrect negative sign in claims system	0.01%	Applied absolute value - magnitude is correct, only the sign is wrong.
total_claim_amount_12m	Outlier: Values above \$400,000 - catastrophic claims (ICU, major surgery); legitimate but extreme	0.09%	Retained values; created high_claim_flag = 1 as indicator for model to handle separately.
late_payment_count_12m	Outlier: Values above 12 - physically impossible in a 12-month window; wrong field pasted in batch upload	0.08%	Capped at maximum valid value of 12.
state	Inconsistent state names - mix of full state names, 2-letter abbreviations.	1.68%	Mapped 51 canonical full state names (50 states + DC) using standardisation dictionary
education_level	Inconsistent education level - abbreviations and alternate spelling.	1.45%	Mapped all variants to 5 canonical values: High School, Some College, Bachelor's, Master's, PhD/Professional.
employment_status	Inconsistent employment status.	1.33%	Mapped all variants to 6 canonical values: Full-time, Part-time, Self-employed, Retired, Student, Unemployed.
plan_tier	Inconsistent plan tier.	0.81%	Mapped all variants to 4 canonical values: Bronze, Silver, Gold, Platinum.
payment_frequency	Inconsistent payment frequency.	1.01%	Mapped all variants to 3 canonical values: Monthly, Quarterly, Annual
policy_type	Inconsistent policy type.	79.67%	Mapped all variants to 7 canonical policy type names using standardisation dictionary
churn_type	Inconsistent formats.	1.20%	Mapped all variants to 8 canonical churn type values
All columns	Duplicate rows.	0.06%	Dropped duplicates.
customer_id	Duplicate customer_ids.	0.30%	Dropped duplicate customer ids and row with fewer null rows per customer_id was retained.

multi_policy_flag	Logical inconsistency - multi_policy_flag=1 but num_policies=1.	0.24%	Updated multi_policy_flag=0 wherever num_policies=1.
subsidy_amount_monthly	Logical inconsistency - subsidy_eligible=0 but subsidy_amount_monthly>0.	0.18%	Updated subsidy_amount_monthly=0 wherever subsidy_eligible=0.
complaint_resolution_days	Logical inconsistency - complaint_flag=0 but complaint_resolution_days>0	0.16%	Updated complaint_flag=1 wherever complaint_resolution_days>0.
churn_type	Logical inconsistency — churn_flag=0 (retained member) but churn_type is not 'No Churn'.	0.14%	Updated churn_type="No Churn" wherever churn_flag=0.
late_payment_count_12m	Logical inconsistency — autopay_enabled=1 but late_payment_count_12m>=4.	0.41%	Left as is — autopay failure is a legitimate business event (card expired, insufficient funds).

EXPLORATORY DATA ANALYSIS

Initial Insights:

Metric / Dimension	Insights
Churn Baseline	24.2% of members lapse annually - 1 in 4. Total CLV at risk exceeds \$750 million.
premium_change_pct	Churned members faced 8.2% average premium increases vs 5.6% for retained - the strongest churn driver across all variables.
multi_policy_flag	Single-policy members churn at 27.0% vs 15.2% for multi-policy - the largest gap of any variable and the strongest retention lever.
autopay_enabled	No-autopay members churn at 29.9% vs 21.6% - 8.3pp gap. 30.6% of members have no autopay, representing an immediately actionable opportunity.
payment_frequency	Monthly payers churn at 25.5% vs 21.8% annually. 63.7% of members pay monthly - the majority are in the highest-risk payment category.
quote_requested_flag	Members who requested a competitor quote churn at 33.9% vs 22.9% - strongest early warning signal detectable in the CRM.
complaint_flag	Members who filed a complaint churn at 33.4% vs 23.7% - service issues directly preceded lapse.
coverage_downgrade_flag	Members who downgraded coverage churn at 31.2% vs 23.3% - financial stress precedes lapse.
plan_tier	Churn decreases from Bronze (26.0%) to Platinum (20.9%). Silver is the largest segment at 34.7% of members.
employment_status	Students (30.5%) and Unemployed (26.8%) churn most driven by financial instability. Full-time employed (49.1% of members) are the most stable.

annual_income	Churned members earn 3,560 less on average - lower income is a consistent financial pressure signal.
customer_tenure_months	Churned members have 4.9 fewer months of average tenure - the first year is the highest-risk window.
rfm_segment	Potential Loyalists is the largest segment (38.6%) at 26.3% churn. Champions (13.8%) churn at only 17.5% - a 12.2pp spread confirms RFM effectively separates members by loyalty.
age_band	Churn decreases steadily with age - youngest members (18–24) churn at 28.1%, dropping to 22.5% for members aged 75+. Younger members combine financial instability with lower plan attachment, making them a higher-risk group despite being a smaller share of the base.
education_level - churn	Churn decreases as education level increases - High School members churn at 26.1% while PhD/Professional members churn at 20.7%. However, the differences are modest, and education is a weaker churn predictor than plan tier or payment frequency.
education_level - CLV	Average CLV increases clearly with education level from High School (5,028) to PhD/Professional (8,927) - a 3,899 gap. Higher education correlates with higher income, higher plan tier selection, and significantly higher lifetime value.
num_policies	Retained members hold 1.68 policies on average vs 1.37 for churned - a 0.31 difference confirming that deeper product engagement predicts loyalty.
Multicollinearity flag	monthly_premium and customer_lifetime_value are highly correlated ($r = 0.74$) - do not include both in the same regression model.
Not significant	credit_score, num_claims_12m, rx_monthly_cost, loss_ratio_12m, and us_region show negligible differences between churned and retained — not meaningful churn predictors.

Deep Dive Insights:

Metric / Dimension	Insights
Plan Tier × Payment Frequency	Monthly payment produces the highest churn within every plan tier. Bronze + Monthly is the highest risk combination and Platinum + Annual is the lowest. The 3–4pp monthly penalty is consistent across all tiers - payment frequency is an independent churn driver regardless of plan type.
Employment × Plan Tier	Students churn at 27–32% across all plan tiers — even on Platinum. Upgrading plan tier alone will not retain Students or Unemployed members - the root cause is employment situation, not plan affordability.
Premium Change % by Plan Tier	The average premium increase is almost identical across all tiers at ~6.2%. Churned members faced ~8.2% increases vs ~5.6% for retained - a consistent 2.6pp gap in every tier. Premium increase caps need to apply across all plan tiers, not just Bronze.

Quote Requested Flag	11.2% of members requested a competitor quote and churn at 33.9% vs 22.9% for those who did not. The churn spike from competitor shopping occurs in every plan tier and employment group. Tenure is identical between the two groups - competitor shopping is not a new-member behavior.
Tenure Band × RFM Segment	Churn decreases with tenure across every RFM segment — the first year is the highest-risk window in all segments. Lost/Price-Sensitive members remain above 25% churn even at the longest tenures — time alone cannot retain this segment. Potential Loyalists show the most improvement with tenure - the most convertible segment with the right early intervention.
Tenure Band - Age vs Tenure Effect	The New band (0–12m) is not dominated by younger members — 55–64 is actually the largest age group at 20.4%, with 18–24 making up only 6.1%. However, 18–24 members churn at 30.8% inside the new band vs 28.1% overall - slightly elevated but not dramatically different. This confirms the high first-year churn is a genuine tenure effect, not an age effect. Younger members churn more because of financial instability, not simply because they are new.
Age Band × Plan Tier	Churn decreases with age within every plan tier — the pattern is consistent regardless of plan type. Even Platinum members aged 18–24 churn at 24.8% - well above Platinum members aged 75+ at 19.7%. Age is an independent churn driver that operates across all plan tiers, not just cheaper ones.
Behavior Flags	Multi-policy membership reduces churn by 11.8pp - the single strongest retention lever. Autopay reduces churn by 8.3pp at near-zero cost - highest ROI action available. Over a third of churned members did not have autopay enabled. 15.7% of churned members had already requested a competitor quote — a missed intervention opportunity.
NPS Score	Detractors (NPS 0–4) churn at 34.5% - 10pp above average — but 99.9% of them also have <code>complaint_flag = 1</code> . NPS low scores and complaint filing are almost perfectly overlapping - NPS adds no independent predictive value beyond what the complaint flag already captures. Only 1.4% of members are Detractors, further limiting its usefulness as a standalone signal.

HYPOTHESIS TESTING

Results Summary:

Seven formal hypotheses were tested at $\alpha = 0.05$. All seven were rejected. Effect sizes are reported alongside p-values because with $n = 500,000$, virtually any difference will be statistically significant — effect size determines practical importance.

#	Hypothesis	Test	Statistic	p-value	Effect Size	Decision
H1	Churn differs by plan tier	Chi-Square (df=3)	$\chi^2=637.70$	<0.001	V=0.036 Small	Reject H ₀
H2	Premium differs churned vs retained	Welch t-test	t=-19.74	<0.001	d=-0.065 Small	Reject H ₀
H3	CLV differs by education level	One-Way ANOVA	F=13,540	<0.001	$\eta^2=0.098$ Large	Reject H ₀
H4	Churn differs by payment frequency	Chi-Square (df=2)	$\chi^2=832.83$	<0.001	V=0.041 Small	Reject H ₀
H5	Premium change % higher for churned	Welch t-test	t=97.52	<0.001	d=0.329 Medium	Reject H ₀
H6	Churn differs by age band	Chi-Square (df=6)	$\chi^2=414.58$	<0.001	V=0.029 Small	Reject H ₀
H7	Churn lower for multi-policy members	Chi-Square (df=1)	$\chi^2=6,981$	<0.001	V=0.118 Medium	Reject H ₀

Key Interpretation:

- **H5 - Most Actionable:** H5 (premium change) has the largest practical effect among continuous variables (Cohen's d = 0.329 - medium). Churned members faced 8.20% average increases vs 5.56% for retained - a 2.64pp gap that is both statistically and practically significant. This is the most actionable finding in the hypothesis testing section.
- **H7 - Strongest Structural Finding:** H7 (multi-policy) has the largest effect size of all seven Chi-Square tests (Cramér's V = 0.118 - medium). The 11.9pp churn reduction for bundled members is the strongest structural retention finding in the entire project.
- **H3 - CLV Driver:** H3 (CLV by education) has the largest overall effect ($\eta^2 = 0.098$). CLV increases from \$5,029 (High School) to \$8,929 (PhD/Professional) - a \$3,900 gap driven by income-to-plan-tier correlation. Education predicts CLV more than churn.
- **H1/H2/H4/H6 - Small Effects:** H1, H2, H4, and H6 are statistically significant but show small effect sizes. The differences are real but modest in practical terms - do not over-prioritize these variables in retention strategy design.

RFM SEGMENTATION & FACTOR ANALYSIS

RFM Segment Profiles:

Segment	Count	Churn Rate	Avg CLV	Avg Net Premium	Strategy
Champions	68,865	17.0%	\$9,028	\$1,067	Loyalty program · annual renewal incentive · concierge service
Loyal Customers	168,046	22.0%	\$7,119	\$830	Multi-policy cross-sell · autopay enrolment push
Potential Loyalists	193,125	26.0%	\$5,381	\$623	Bundle offer at onboarding · 6-month milestone outreach
At-Risk	47,420	28.0%	\$3,996	\$462	URGENT: personal outreach + competitive price match
Lost/Price-Sensitive	22,544	30.0%	\$3,097	\$358	Subsidy check · plan upgrade offers · hardship program.

- **Biggest Opportunity:** Potential Loyalists is the largest segment (38.6% of members, 193,125 members) with a 26.0% churn rate. This segment represents the biggest retention opportunity by volume - even a modest 5% churn reduction would retain 9,656 additional members annually.
- **Segment Differentiation:** Champions have a 12.2pp lower churn rate than Lost/Price-Sensitive members - confirming RFM scoring successfully separates members by loyalty depth and financial engagement.
- **Lifecycle Insight:** Tenure band analysis confirms churn decreases with tenure in every segment - but Lost/Price-Sensitive remains above 25% even at 15 years of membership. Time alone cannot solve churn for this segment.

Factor Analysis (PCA):

PCA was applied to 15 financial and behavioral variables. Five factors with eigenvalue above 1.0 (Kaiser criterion) were retained, explaining 62.4% of total variance.

Factor	Eigenvalue	Variance %	Cumulative %	Top Variables	Interpretation
F1	3.248	21.66%	21.66%	deductible, out_of_pocket_max, net_monthly_premium	Plan Cost Structure
F2	1.972	13.15%	34.80%	total_claim_amount, loss_ratio, num_claims	Claims Activity
F3	1.607	10.71%	45.51%	monthly_premium, credit_score, net_monthly_premium	Financial Capacity
F4	1.477	9.84%	55.36%	annual_income, digital_engagement, late_payments	Financial Behavior
F5	1.061	7.07%	62.43%	customer_tenure_months, num_policies	Member Loyalty Depth

F5 (Loyalty Depth) is the most strategically relevant factor — it loads on customer_tenure_months and num_policies, directly corresponding to the multi-policy and tenure findings from EDA and hypothesis testing. Members scoring high on F5 are the Champions and Loyal Customers segments.

CHURN CLASSIFICATION MODEL

Model Comparison:

Four classification models were trained on a 100,000-member sample (80,000 train / 20,000 test, 76 features) with a 24.1% churn rate. ROC-AUC was used as the primary metric because accuracy is misleading at 24:76 class distribution.

Model	Accuracy	F1 Score	ROC-AUC	Notes
Logistic Regression	0.609	0.429	0.652	Strong baseline; interpretable; class_weight=balanced
Random Forest	0.759	0.001	0.657	High accuracy but near-zero F1 on churned class
Gradient Boosting	0.760	0.084	0.662	Competitive; lr=0.1, max_depth=4, 50 estimators
XGBoost (Best)	0.760	0.075	0.663	Highest AUC; selected as production model

Best Model — XGBoost Classification Report:

Class	Precision	Recall	F1-Score	Support
Retained	0.76	0.99	0.86	15,177
Churned	0.52	0.04	0.08	4,823
Accuracy			0.76	20,000

- The model correctly identifies 99% of retained members but only 4% of churned members at the default threshold of 0.5. This conservative behavior is by design - at 0.5, the model has high precision (52%) for churned predictions, meaning most flagged members are genuine churn risks.
- Lowering the decision threshold from 0.5 to 0.3 would increase churned recall significantly at the cost of more false positives. For a retention campaign context, a lower threshold is preferable - the cost of missing a churning exceeds the cost of a false-positive outreach call.

Top Features Importances:

Rank	Feature	Importance	Connection to Earlier Findings
1	multi_policy_flag	0.172	Confirms H7 - strongest structural protective factor
2	premium_change_pct	0.165	Confirms H5 - strongest numeric churn driver

3	autopay_enabled	0.061	Confirms EDA behavior flags (8.3pp gap)
4	num_policies	0.050	Supports multi_policy finding - policies count matters
5	quote_requested_flag	0.040	Confirms EDA deep dive - detectable early warning
6	coverage_downgrade_flag	0.030	Confirms EDA - financial stress precedes lapse
7	complaint_flag	0.022	Confirms EDA - service issues predict lapse
8	net_monthly_premium	0.021	Absolute premium level matters after subsidy
9	payment_frequency_Monthly	0.021	Confirms H4 - monthly payment is structural risk
10	num_price_increases_last_3y	0.021	Cumulative pricing history, not just current year

CUSTOMER LIFETIME VALUE (CLV) REGRESSION

Model Performance:

Multiple Linear Regression was applied to predict CLV across all 500,000 members. monthly_premium was excluded due to confirmed multicollinearity with CLV ($r = 0.74$, identified in the EDA correlation heatmap). Age band features were added following the EDA finding that age is an independent CLV driver.

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Top CLV Drivers:

Coefficients show the dollar change in predicted CLV for each unit change in the feature, holding all other variables constant.

Feature	Coefficient	Interpretation
plan_tier_Platinum	+\$1,930	Platinum members generate \$1,930 more CLV than Bronze — plan tier is the strongest driver after excluding premium
plan_tier_Gold	+\$997	Gold members generate \$997 more CLV than Bronze
us_region_Northeast	+\$895	Higher cost-of-living markets generate higher premiums and CLV
employment_status_Student	+\$892	Unexpectedly positive — students on parent plans may carry high base premiums
age_band_45-54	+\$810	Peak earning years — highest plan tier selection and age-rated premiums
age_band_55-64	+\$723	Age-rated premiums significantly higher in this band
us_region_West	+\$709	Premium-priced market — consistent with Northeast finding
employment_status_Unemployed	-\$472	Unemployed members on cheapest available plans
education_level_High School	-\$368	Lower education correlates with lower income and cheaper plan selection
employment_status_Retired	-\$260	Fixed income limits plan tier selection

Diagnostic Assessment:

- **Residuals vs Fitted:** Fan-shaped pattern (heteroscedasticity) - residuals spread wider as fitted values increase. The model is less precise for high-CLV members. Expected for right-skewed targets.
- **Q-Q Plot:** Points deviate at both tails - residuals are not perfectly normal, driven by the right-skewed CLV distribution. Acceptable for $n = 500,000$ - the Central Limit Theorem ensures coefficient validity.
- **Actual vs Predicted:** Points cluster tightly around the diagonal for CLV up to \$15,000. The model systematically underpredicts members above \$15,000 - a known limitation of OLS on right-skewed targets.

LIFECYCLE ANALYSIS & CHURN FORECASTING

Approach:

The dataset is a single cross-sectional snapshot, not a longitudinal panel. The churn lifecycle was reconstructed using customer_tenure_months (1–180 months) as a pseudo-time axis. Exponential Smoothing with additive trend (statsmodels) was applied for the 24-month forecast. ADF stationarity test was conducted to validate the modelling approach.

Tenure Band Analysis:

Band	Tenure	Members	Churn Rate	Avg CLV	Intervention
New	0–12m	118,580	26.5%	\$7,317	Autopay setup + bundle offer at 90 days
Early	13–36m	150,688	25.5%	\$6,880	Bundle cross-sell at 18-month anniversary
Established	37–84m	146,241	22.9%	\$6,010	Loyalty discount at 3-year anniversary
Loyal	85–120m	44,579	21.5%	\$4,754	VIP program + dedicated account manager
Champion	121–180m	39,912	19.8%	\$3,040	Annual concierge renewal + early-bird discount

- The first year is the highest risk window at 26.5% - 2.4pp above the overall average. Early onboarding interventions (autopay enrolment, bundle offers) have the highest concentration of at-risk members.
- Avg CLV decreases with tenure because CLV is forward-looking - newer members have more remaining lifetime value ahead of them, not because they are less valuable.
- A 6.7pp improvement from New (26.5%) to Champion (19.8%) confirms that retention investment in the first 3 years yields the most measurable lifecycle improvement.

Stationarity & Forecasting:

The ADF test returned $\text{stat} = -2.071$, $p = 0.256$ - the series is NOT stationary. This means the churn rate follows a declining trend over tenure rather than fluctuating around a fixed mean. Exponential Smoothing with additive trend is the appropriate model because it explicitly captures and projects the downward trajectory.

Period	Forecast	95% CI Lower	95% CI Upper
Month 181 (start)	19.46%	14.73%	24.18%
Month 192 (mid)	19.26%	14.53%	23.98%
Month 204 (end)	19.04%	14.32%	23.77%

- The forecast projects a very gradual decline of only 0.42pp over 24 months - the downward trend continues but flattens considerably, suggesting churns is approaching a natural floor.
- The 95% CI is $\pm 4.7\text{pp}$ - wider than the projected decline itself. In a pessimistic scenario, churns could remain at current levels rather than declining.
- Even at 15+ years of membership, churn never approaches zero. Time alone does not solve churn - active structural interventions remain necessary throughout the entire member lifecycle.

CONCLUSIONS & LIMITATIONS

Consolidated Findings:

Finding	Evidence	Actionability
Premium increases are the #1 controllable driver	H5: Cohen's $d=0.329$ · Model: 2nd feature importance · EDA: 8.2% vs 5.6%	HIGH - directly adjustable by pricing team
Multi-policy bundling is the #1 protection lever	H7: $V=0.118$ · Model: #1 feature importance · EDA: 11.9pp gap	HIGH - cross-sell campaign, onboarding flow
Autopay is highest ROI retention action	EDA: 8.3pp gap · 30.6% members without autopay	HIGH - onboarding change only, near-zero cost
Early warning signals are CRM-detectable	Quote flag: 11.0pp · Complaint: 10.1pp · Downgrade: 7.8pp	HIGH - CRM integration of model scores
First year is the highest-risk lifecycle window	Tenure bands: 26.5% New vs 19.8% Champion	HIGH - focused onboarding program
Age drives churn independently across all tiers	H6: significant · EDA: 28.1% (18-24) vs 22.5% (75+)	MEDIUM - age-specific product accessibility
Lost/Price-Sensitive cannot be retained by time	Tenure $\times$ RFM: 25%+ even at 15 years	MEDIUM - price intervention or graceful exit

Limitations:

- The dataset is synthetic - all models must be validated against live production data before CRM deployment.

- Churn is modelled as binary. Plan downgrades (partial lapse) vs full policy cancellation are not distinguished.
- No competitor pricing data - the Competitive Switch churn type (11.5% of churned members) cannot be fully modelled from internal data alone.
- The lifecycle curve is cross-sectional, not longitudinal - true cohort tracking requires time-series panel data.
- XGBoost recall for churned members is only 4% at the default threshold. Threshold tuning and A/B testing of the intervention program are required before full deployment.

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